Nuclear Operating Company

September 11, 2025
NOC-AE-25004126
10 CFR 50.73
STI: 35773211

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

South Texas Project
Units 1 and 2
Docket Nos. STN 50-498 and STN 50-499
Licensee Event Report 2024-004-01
Supplement to Loss of Offsite Power Resulting in Unit 1 Automatic Reactor Trip and Actuation of Emergency Diesel Generators and Auxiliary Feedwater Pumps

Reference: Letter; J. Tomlinson (STP) to Document Control Desk (NRC) – "Licensee Event Report 2024-004-00, Loss of Offsite Power Resulting in Unit 1 Automatic Reactor Trip and Actuation of Emergency Diesel Generators and Auxiliary Feedwater Pumps," September 19, 2024, (NOC-AE-24004064), (ML24263A145)

On September 19, 2024, STP Nuclear Operating Company (STPNOC) submitted the referenced Licensee Event Report. This letter is a supplement to the report to provide updates as a result of the causal investigation. The updated information is denoted by revision bars located in the right-hand margin. The report is submitted in accordance with 10 CFR 50.73.

The event did not have an adverse effect on the health and safety of the public.

There are no commitments in this submittal.

If there are any questions regarding this submittal, please contact Chris Warren at (361) 972-7293 or me at (361) 972-7566.

Robert B. Norris
Site Vice President

Attachment: Units 1 and 2 LER 2024-004-01, Supplement to Loss of Offsite Power Resulting in Unit 1 Automatic Reactor Trip and Actuation of Emergency Diesel Generators and Auxiliary Feedwater Pumps

cc:
Regional Administrator, Region IV
U.S. Nuclear Regulatory Commission
1600 E. Lamar Boulevard
Arlington, TX 76011-4511

Attachment

Units 1 and 2 LER 2024-004-01

Units 1 and 2 LER 2024-004-01, Supplement to Loss of Offsite Power Resulting in Unit 1 Automatic Reactor Trip and Actuation of Emergency Diesel Generators and Auxiliary Feedwater Pumps

NRC FORM 366 (04-02-2024) U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
South Texas Project Unit 1	■ 050 / ☐ 052	00498	1 OF 5

4. Title
Loss of Offsite Power Resulting in Unit 1 Automatic Reactor Trip and Actuation of Emergency Diesel Generators and Auxiliary Feedwater Pumps

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name		Docket Number
07	24	2024	2024	- 004 -	01	09	11	2025	South Texas Project Unit 2	■ 050	00499
									N/A	☐ 052	

9. Operating Mode	10. Power Level
1	100%

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	✓ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Chris Warren, Licensing Engineer	361-972-7293

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	N/A	INS	G080	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
✓ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On July 24, 2024, STP Unit 1 experienced a Loss of Offsite Power (LOOP), resulting in an automatic reactor trip and actuation of all three Unit 1 Standby Diesel Generators (SBDG) and all four Auxiliary Feedwater (AFW) pumps. All three trains of Engineered Safety Feature (ESF) buses were energized and all equipment responded as expected without any complications with the exception of Steam Generator Power Operated Relief Valve "1C." Unit 2 experienced a partial LOOP and automatic actuation of SBDG 22 and one of four AFW pumps. The systems for each unit actuated due to the Switchyard experiencing electrical faults during the Shunt Reactor RT-2 fire event, which was caused by the insulation breakdown on the C-Phase (H3) GE Type U high-voltage bushing. Immediate corrective actions included restoring offsite power circuits. A Transmission Distribution Service Provider (TDSP) had not established and implemented adequate management controls governing design changes to equipment in the Switchyard to ensure that changes were consistent with design requirements and aligned with requirements and recommendations of applicable Industry Standards; and subsequently, STPNOC did not rigorously implement management controls during the review and approval of design changes developed by the TDSP. Planned corrective actions include enhancements to STP procedure 0PGP05-ZA-0002, "10CFR50.59 Evaluations" and development and implementation of written instructions that include management controls to govern the review and approval of all four TDSPs' Switchyard changes.

NARRATIVE

1. Description of the Reportable Event

A. Reportable Event Classification

This event is reportable pursuant to 10 CFR 50.73(a)(2)(iv)(A) as an event that resulted in the valid actuation of the Reactor Protection System (RPS) including reactor trip (10 CFR 50.73(a)(2)(iv)(B)(1)), and automatic actuation of all four Unit 1 Auxiliary Feedwater (AFW) Pumps and Unit 2 AFW Pump 22 (10 CFR 50.73(a)(2)(iv)(B)(6), and automatic actuation of all three Unit 1 Standby Diesel Generators (SBDG) and Unit 2 SBDG 22 (10 CFR 50.73(a)(2)(iv)(B)(8)).

B. Plant Operating Conditions Prior to Event

Prior to the event, Units 1 and 2 were in Mode 1 at 100% power.

C. Status of Structures, Systems, and Components That Were Inoperable at the Start of the Event that Contributed to the Event

At the start of the event, there were no structures, systems, or components (SSCs) that were inoperable that contributed to the event.

D. Narrative Summary of the Event

Timeline (Note: All times are in Central Daylight Time)

Unit 1 - 07/24/2024 (0702) - Unit 1 reactor trip due to fire in the switchyard. The 345kV North Bus and the Unit 1 Generator tripped. All SBDGs automatically actuated and sequenced on a Loss of Offsite Power. The following busses were deenergized: 13.8kV Auxiliary busses 1F, 1G, 1H, and 1J, 13.8kV Standby busses 1F, 1H, and 1G, and 480V Load Center 1W.

Units 1 and 2 - 07/24/2024 (0702) - Due to loss of two independent offsite power circuits, entered Technical Specification 3.8.1.1, Action 'e'.

Unit 1 - 07/24/2024 (0702) - All four Reactor Coolant Pumps (RCP) lost power. Entered Technical Specification 3.4.1.2.

Unit 2 - 07/24/2024 (0703) - Report of explosion and fire in the switchyard. Entered off-normal procedure 0POP04-ZO-0008, Fire/Explosion.

Unit 1 - 07/24/2024 (0735) - Verified natural circulation with Steam Generator (S/G) Power Operated Relief Valves (PORV) open in Manual.

07/24/2024 (0800) - Bay City Fire Department arrive on site at the switchyard.

Unit 1 - 07/24/2024 (0848) - Spent Fuel Pool pump 1B is running, restoring cooling to the Spent Fuel Pool.

07/24/2024 (0925) - Fire in Shunt Reactor RT-2 is extinguished.

NARRATIVE

D. Narrative Summary of the Event (continued)

Timeline (Note: All times are in Central Daylight Time)

Unit 1 - 07/24/2024 (0953) - S/G PORV 1C declared inoperable and nonfunctional due to not operating in AUTO or MANUAL from the Control Room.

Unit 1 - 07/24/2024 (1054) - Completed NRC 4-hour Non-Emergency Notification under 10 CFR 50.72(b)(2)(iv)(B) for a Reactor Trip from 100% power (RPS actuation). Event Notification Number 57237.

Unit 1 - 07/24/2024 (1346) - Started Reactor Coolant Pump 1D. Reactor Coolant Loop D and its associated steam generator and reactor coolant pump are in operation with reactor trip breakers OPEN. Exited LCO 3.4.1.2, Action 'c'.

Unit 2 - 07/24/2024 (1350) - Both 345kV North and South buses are OPERABLE and all three ESF buses are powered by the Unit Auxiliary Transformer. Exited Technical Specification 3.8.1.1, Action 'e' and entered Action 'a.'

07/24/2024 (1455) - Completed NRC 8-hour non-emergency notification under 10 CFR 50.72(b)(3)(iv)(A) as an event that resulted in the valid actuation of a PWR auxiliary system and emergency AC electrical power system. Completed NRC 4-hour non-emergency notification under 10 CFR 50.72(b)(2)(xi) for the issuance of a news release. Event Notification EN# 57237.

Unit 2 - 07/24/2024 (1537) - Normal offsite power to Unit 2 restored. Exited Technical Specification 3.8.1.1, Action 'a.'

Unit 1 - 07/25/2024 (0057) - Switchyard Breakers Y0520 and Y0510 closed. Main and Auxiliary transformers are energized.

Unit 1 - 07/25/2024 (0933) - Normal offsite power to Unit 1 restored. Exited Technical Specification 3.8.1.1, Action 'a'.

E. Method of Discovery

The event was self-revealing when Unit 1 experienced an automatic reactor trip coincident with a Loss of Offsite Power and actuation of the SBDGs for all three ESF buses.

II. Component Failures

A. Failure Mode, Mechanism, and Effects of Failed Components

Two independent failure analyses were performed and concluded that the Shunt Reactor RT-2 H3 GE Type U high-voltage bushing failed due to an insulation breakdown. The independent failure analyses could not determine the cause of the insulation breakdown due to extensive damage sustained during the event. The failure of the bushing resulted in the subsequent fire. The smoke plume reached the A-phase conductor associated with the Unit 1 Main Generator Transformer and the C-phase conductor associated with the No. 1 Standby Transformer, allowing phase-to-ground faults through the plume and the loss of two independent offsite transmission circuits.

NARRATIVE

B. Cause of Component Failure

The independent failure analyses could not determine the cause of the insulation breakdown due to extensive damage sustained during the event.

C. Systems or Secondary Functions That Were Affected by the Failure of Components with Multiple Functions

Standby Transformer 1 was disconnected from the 345kV North Bus due to collateral damage to a tower, insulators, and overhead conductors.

D. Failed Component Information

System: N/A - Shunt Reactor RT-2 is owned, operated, and maintained by a partnership of four Electrical Transmission Utility Companies
Component: Insulator {INS}
Manufacturer: General Electric {G080}
Model: {GE 11B877BB}

III. Analysis of Event

A. Safety System Responses that Occurred

The Unit 1 Reactor Protection System actuated, automatically tripping the Unit 1 reactor following the main generator lockout. All three SBDGs automatically started within the required time frame and provided power to all three Engineered Safety Feature (ESF) trains. All four AFW pumps started, providing adequate cooling for the steam generators during cooldown following the reactor trip.

The Unit 2 SBDG 22 automatically started within the required timeframe, providing power to ESF Train 'B'. AFW Pump 22 automatically started, ensuring adequate cooling for the steam generators.

B. Duration of the Safety System Inoperability

Not applicable. No safety systems that contributed to this event were inoperable.

C. Safety Consequences and Implications

All nuclear safety systems for Units 1 and 2 responded as expected without any complications except for Unit 1 S/G PORV ‘1C’. There were no inoperable safety systems that contributed to the event and no safety systems were rendered inoperable as a result of the event. There were no actual consequences to the health and safety of the public as a result of this event.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
South Texas Project Unit 1	■ 050 / ☐ 052	00498	2024	- 004	- 01

NARRATIVE

IV. Cause of Event

A Transmission Distribution Service Provider (TDSP) had not established and implemented adequate management controls governing design changes to equipment in the Switchyard to ensure that changes were consistent with design requirements and aligned with requirements and recommendations of applicable Industry Standards; and subsequently, STPNOC did not rigorously implement management controls during the review and approval of design changes developed by the TDSP. As a result, in 2014 the Shunt Reactors were relocated, and their associated circuit switchers were replaced with circuit breakers without adequately evaluating the change. Consequently, due to the relocation of Shunt Reactor 2, on July 24, 2024, a single bushing failure caused a fire that resulted in loss of the independent transmission circuits, which caused a loss of offsite power to Unit 1, and resulted in a partial loss of offsite power to Unit 2.

V. Corrective Actions

The following Corrective Actions are planned and tracked in the STP Corrective Action Program:

1. Revise STP procedure 0PGP05-ZA-0002, "10CFR50.59 Evaluations," to require 10CFR 50.59 Screenings and Evaluations address all elements of the change; and change the Usage category to "Referenced" as delineated in STP Procedure 0PAP01-ZA-0101, “Plant Procedure Writer’s Guide.”

2. Develop and implement written instructions that include management controls to govern the review and approval of Transmission Distribution Service Providers' procedure changes, design changes, tests or changes in the conduct of other activities which might affect compliance with regulatory requirements, the Updated Final Safety Analysis Report and/or NRC commitments involving the Switchyard and associated transmission lines and equipment which could affect offsite power supply to STP consistent with the South Texas Project Interconnection Agreement, Exhibit D, Section V, “Review and Approval.”

VI. Previous Similar Events

A review of events over the past 3 years did not identify any similar events with the same cause as this event.